

Global Renewable Energy in 2025

How Far Have We Come?

A data-driven overview of renewable energy's growing share in the global energy and electricity mix, spanning hydropower, wind, and solar technologies.

Data: Our World in Data | Ember (2026) | Energy Institute – Statistical Review of World Energy (2025)

75% of Global GHG Emissions Still Come from Fossil Fuels

75%

of global greenhouse gas
emissions from fossil fuels

5M+

premature deaths per year
from fossil fuel air pollution

150+

years of fossil fuel
dominance since Industrial
Revolution

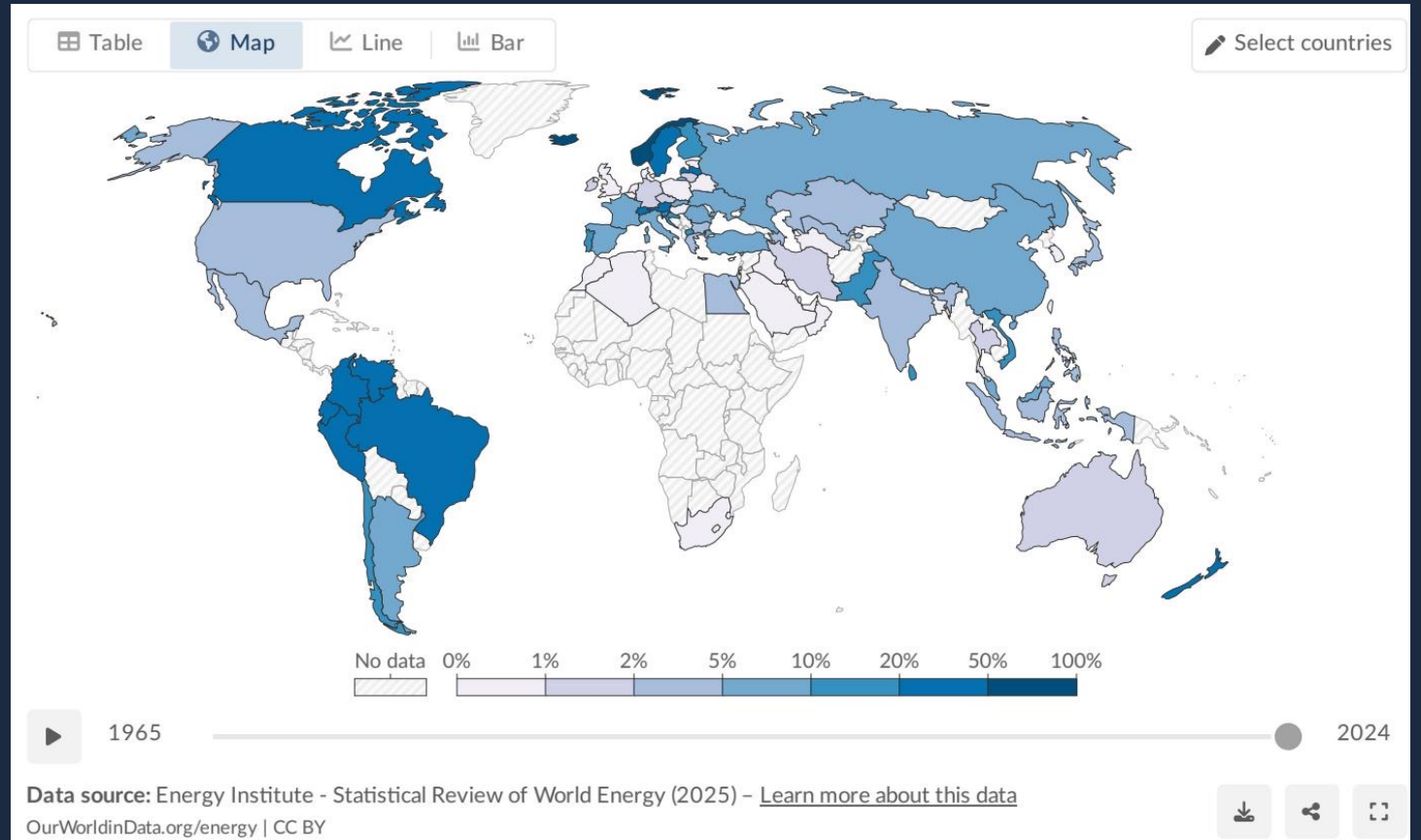
Since the Industrial Revolution, most countries' energy systems have been dominated by fossil fuels. Three-quarters of global greenhouse gas emissions come from burning fossil fuels for energy. Fossil fuel combustion is also responsible for local air pollution that causes at least 5 million premature deaths per year. To reduce CO₂ emissions and air pollution, the world must rapidly shift toward low-carbon sources — nuclear and renewable technologies.

Renewables Now Supply ~14% of Global Primary Energy

~14%

of global primary energy
from renewables (2024)

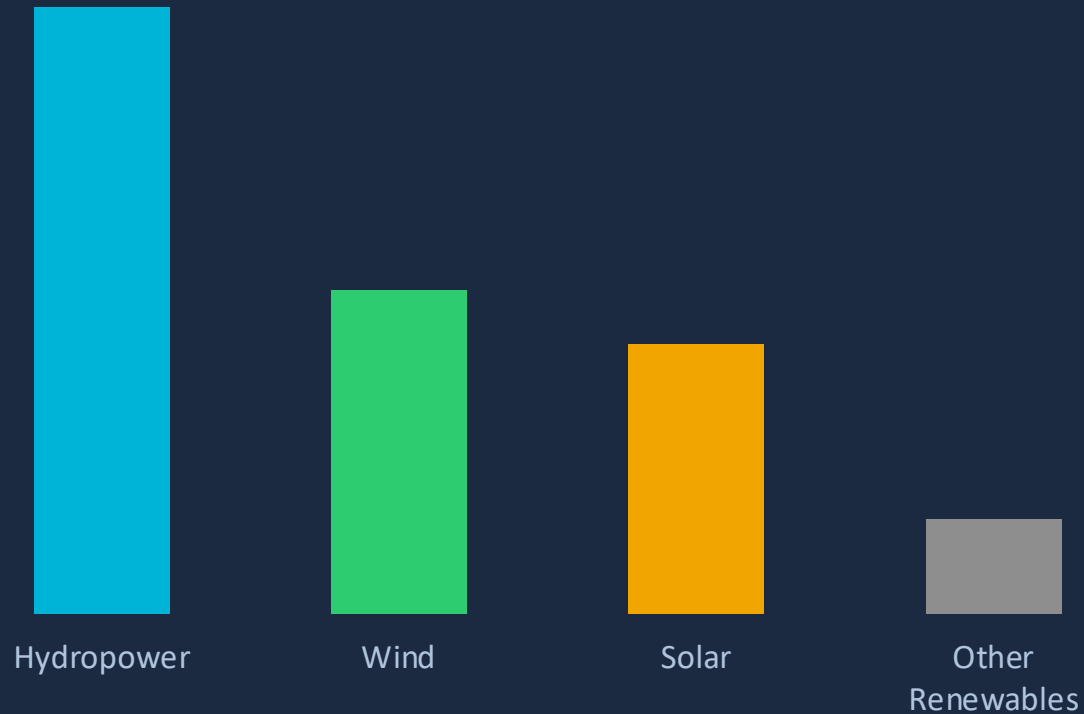
- Primary energy includes electricity, transport, and heating combined
- Transport and heating are harder to decarbonize — more reliant on oil and gas
- Renewable share in total energy is lower than in electricity alone
- Norway, Brazil, and New Zealand lead; Middle East and parts of Asia remain below 5%



Share of primary energy from renewables by country, 2024 (substitution method)

Renewable Electricity Generation Is Growing Rapidly, Led by Wind and Solar

Generation (TWh)



Hydropower

~4,500 TWh | ~47% of renewable electricity

Wind

~2,400 TWh | ~25% of renewable electricity

Solar

~2,000 TWh | ~21% of renewable electricity

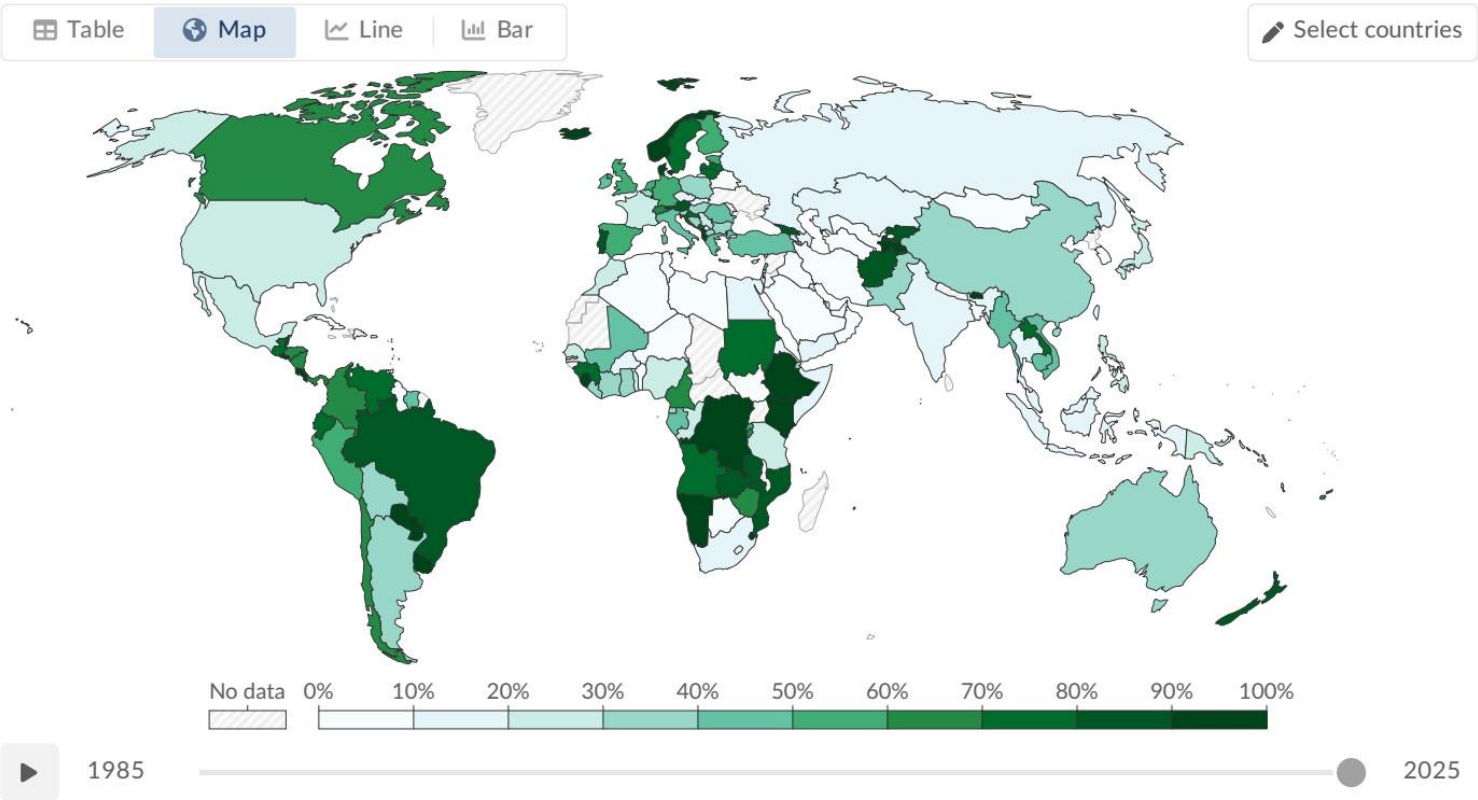
Other

~700 TWh | ~7% of renewable electricity
Wind and solar combined now nearly match hydropower's output and they are growing much faster.

Almost One-Third of Global Electricity Now Comes from Renewables

Share of electricity production from renewables, 2025

Renewables include solar, wind, hydropower, bioenergy, geothermal, wave, and tidal sources.



Data source: Ember (2026); Energy Institute Statistical Review of World Energy (2025); [Learn more about this data](#)
OurWorldinData.org/energy | CC BY

Region / Country	Renewable Share
Norway	~98%
Brazil	~85%
Canada	~65%
Germany	~55%
China	~35%
India	~25%
United States	~25%
World average	~30%
South Africa	~10%
Saudi Arabia	<2%

Several African and Middle Eastern countries remain below 10%.

Hydropower Remains the Largest Renewable Source at ~50% of Generation

~4,500 TWh

global hydropower generation (2024)

~47%

of all renewable electricity from hydro

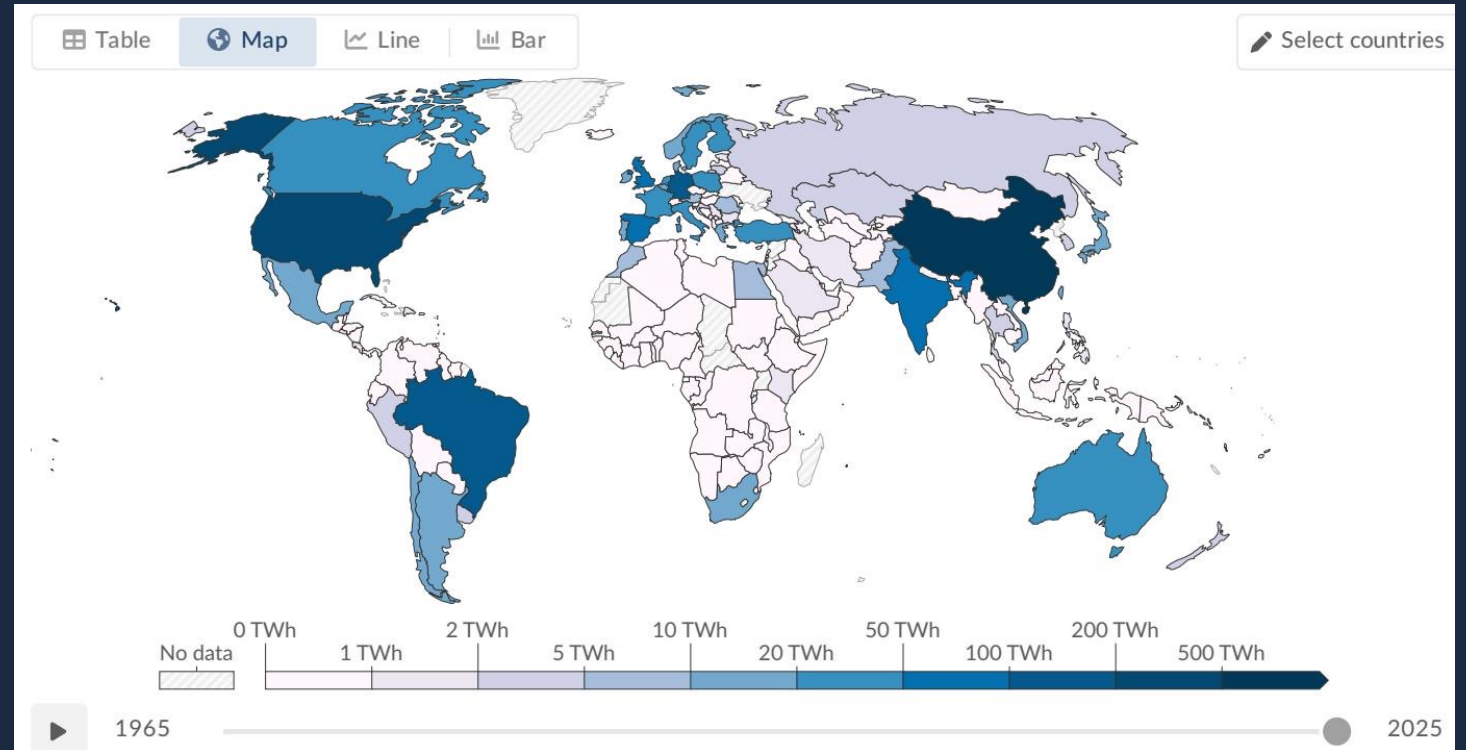
~50%

of renewable generation — still the single largest source

100+ years

of electricity generation at scale

Top producers: China, Brazil, Canada



Hydropower generation by country, 2025

Wind and Solar Are the Fastest-Growing Energy Technologies in History

WIND

~2,400 TWh | ~25% of renewable electricity

Major contributors: Europe, China, US

SOLAR

~2,000 TWh | ~21% of renewable electricity

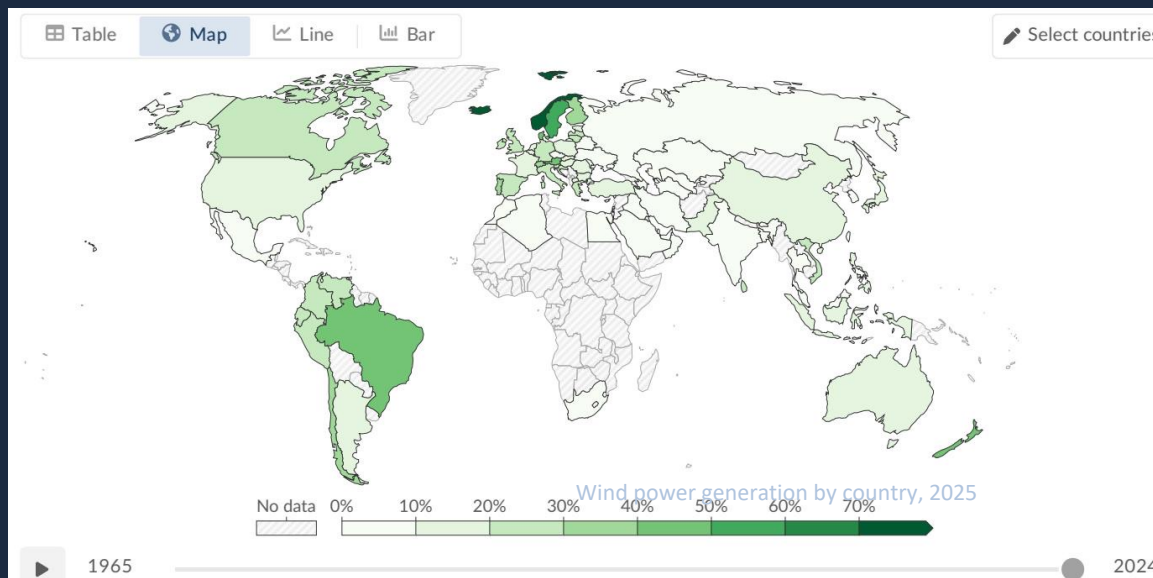
Fastest-growing source; dramatic growth since 2010

COMBINED

~4,400 TWh

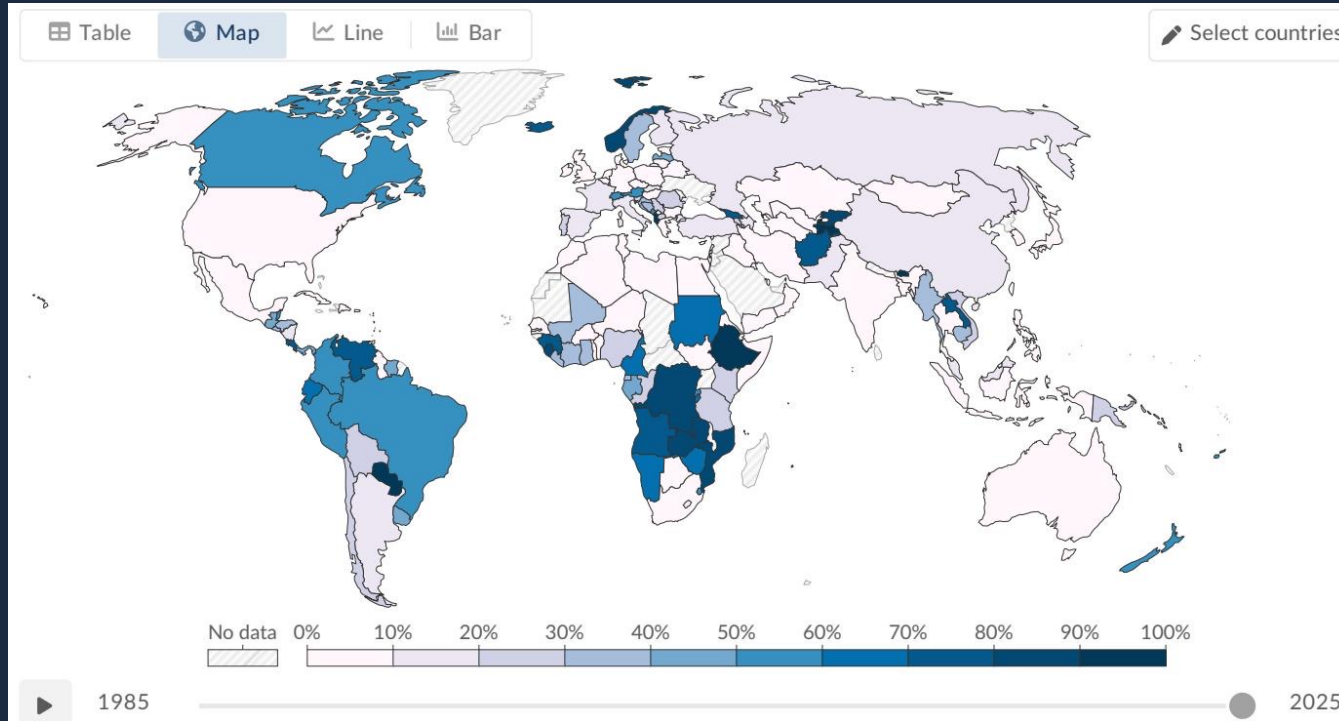
~46% of renewable electricity

Nearly matching hydro output



Wind and solar have seen exponential growth since 2010. Falling costs and strong policy support continue to accelerate deployment worldwide.

Stark Regional Gaps: Sub-Saharan Africa Lags While Scandinavia and Latin America Lead



Scandinavia & Latin America

65–98% renewable electricity — abundant hydropower

Europe (Denmark, Germany)

55%+ renewable electricity — heavy investment in wind & solar

China & India

~25–35% — rapidly scaling renewables but still fossil-dependent

Sub-Saharan Africa

Minimal renewable infrastructure; most countries below 10%

Middle East

Almost entirely fossil-fuel dependent; Saudi Arabia <2%

Share of electricity from hydropower by country, 2025

Key Takeaways: Renewables Are Growing Fast but the Transition Must Accelerate

1

~14% of primary energy, ~30% of electricity from renewables

A significant milestone, but far from sufficient for climate targets. The electricity sector leads; transport and heating lag behind.

2

Hydropower remains dominant but growth is limited

Generating roughly half of all renewable electricity, hydro's expansion potential is constrained compared to wind and solar.

3

Wind and solar are the fastest-growing energy technologies in history

Combined, they now produce nearly as much electricity as hydropower, driven by falling costs and policy support since 2010.

4

Stark regional disparities persist

Scandinavia and Latin America exceed 60–90% renewable electricity, while much of the Middle East and Sub-Saharan Africa remain below 10%.

5

The pace of transition must accelerate

Transport and heating lag far behind electricity in decarbonization. Meeting climate targets requires faster deployment across all sectors and regions.

The energy transition is underway, but current pace is insufficient. Accelerating renewable deployment — especially in lagging sectors and regions — is essential to meet climate goals.